

Space-time analysis of accident frequency and the role of built environment in mitigation

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ABSTRACT

The focus of this paper is to analyze the trends and locations of accidents in the Greater Melbourne Area (GMA) during a 15-year period (2006–2020). The places where accidents were most prevalent were discovered and the reasons which are contributing to the high accident rates in those areas are determined. Analyzing the patterns over time and variations in the frequency of accidents helped to identify areas that have improved or deteriorated in terms of road safety. A Tweedie model was developed to examine the intricate interaction between the accident frequency and its potential contributing factors such as socio-demographics, road transport infrastructure, and built environment. Ultimately, a clustering analysis was performed to elucidate the dispersion of road accident risk ratings among different local government areas (LGAs), offering useful insights into road safety initiatives and prioritization.

1. Background

The present worldwide mean of road fatalities is at 17.4 deaths per 100,000 individuals, with a greater occurrence in lower-income nations and a reduced incidence in higher-income nations (Jamal et al., 2020). However, the severity and resulting consequences of accidents vary significantly among developed nations, influenced by disparities in safety regulations, infrastructure quality, emergency response systems, cultural attitudes, and economic conditions (Adanu et al., 2017). Australia has a fatality rate that is higher than many OECD (Organization for Economic Co-operation and Development) nations and even higher than the average rate for these nations, and its road fatality rate is 1.9 times that of the United Kingdom (BITRE, 2021).

Road safety has been a policy priority in Australia for decades, however the rate of road transport injuries is not declining. Hospitalization rates over the last 9 years have increased, with no decline in ‘high threat to life’ hospitalizations (BITRE, 2021). According to Australian statistical data, it has been observed that there is an estimated annual occurrence of over 1400 deaths in 2020–21 and more than 61,500 hospitalizations in 2021–22 reported injuries resulting from road-related incidents (AIHW, 2022). The average cost of a hospitalization due to a traffic-related injury is AUS \$239,000, putting the yearly

cost of road trauma at an estimated AUS\$29.7 billion (Australian Automobile Association, 2017). These figures impose significant burdens on healthcare systems and communities.

The Australian Federal Government and State Governments have introduced a variety of policies and initiatives to minimise the number of car accidents, including road safety campaigns, infrastructure improvements, speed restrictions, and graded licencing process. In addition, the government has imposed safety regulations for vehicles sold in Australia, as well as sanctions and repercussions for reckless driving (Department of Infrastructure, 2018). The Road Safety Victoria and the Victorian Road Safety Partnership produced the Victorian Road Safety Plan 2021–2030 to eliminate road fatalities in Victoria by 47,2050, beginning with half by 2030. Road safety, practical measures, and future-proofing Victoria are its goals. Short-term action plans, legislation, innovation and technology, infrastructural improvements, public awareness campaigns, education programmes, enforcement, and other methods will execute the approach (TAC, 2021).

The frequency of road accidents in Australia may be affected by a variety of factors, including but not limited to socio-demographics, road design, volume and speed, driver behavior, built environment and regulations enforcement. Personal and family characteristics such as age, income, and gender all affect the frequency of road accidents. These

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qualities impact driving, risk perception, and road safety. Young and inexperienced drivers are more prone to age into risky practices and have poor hazard awareness, increasing accident risk (Borowsky et al., 2010). Inexperienced drivers are prone to accidents because it is hard for them to anticipate and respond to unforeseen circumstances. However, senior drivers' cognitive and physical abilities may impair their driving. Thus, age and accident frequency create a U-shaped relationship, with younger and older drivers at higher risk (Kam, 2003). Personal or family income, vehicle type, maintenance, and safety features may impact accident frequency. Higher-income persons may drive safer vehicles, leading to fewer accidents. Better vehicle maintenance may reduce mechanical failure accidents for higher-income people (Elvik, 2009). Lower-income drivers may drive older cars with fewer safety features, increasing accident risk.

Gender has been a contentious issue in road safety research. Some studies show that men are greater prone to risky driving behavior, including aggressive driving and speeding, leading to a higher accident rate (Constantinou et al., 2011), while others claim that women may be involved in other types of accidents, such as low-speed collisions, due to driving behaviors (Schuller et al., 2000). These modifications may trigger gender-specific accidents. The size of a household might affect road safety (Taubman-Ben-Ari & Katz-Ben-Ami, 2013). Multi-driver families generally have different driving styles. Increased road usage and potentially greater vehicle risk result from this. Conflicting schedules, shared cars, and diverse household destinations may complicate traffic and potentially increase accident risk. The intricate relationship between speed and accident rates has been thoroughly examined, finding unexpected road type classifications. Local road accident rates are greater than larger routes as speed rises (Aarts and Van Schagen, 2006). This phenomenon is intricately regulated by lane width, junction density, and traffic flow, emphasizing the vital role of accurate speed design in tolerating varying speeds and averting accident (Aarts and Van Schagen, 2006). Street network characteristics' role in road safety is unclear.

The built environment can influence accident frequency and severity in different ways (Gálvez-Pérez et al., 2022; Merlin et al., 2020; Lym et al., 2022; Kamel et al., 2019; Zafri and Khan, 2022; Soltani et al., 2023a,b). Several research have elucidated that built environment elements play a significant role in affecting the incidence and severity of vehicle accident. The architecture of street networks, speed dynamics, junction design complexity, land use configurations, and even the presence of specific utilities are all factors. Denser street networks with more intersections per unit area minimise all accident severity levels, however, street connections and vehicle lanes along important arteries enhance accident rates. Low street network density areas are more at risk of fatal or major accident. As intersection density increases, safety increases (Marshall and Garrick, 2011). These results underscore the necessity for measured road design that balances the imperative of maintaining smooth traffic flow with the hazards of increasing connectivity and lane expansion.

The presence of highway intersection lighting is linked to a 12% decrease in the ratio of accident that occur at night compared to during the day. Additionally, increased visual performance resulting from roadway lighting is expected to lead to a decrease in the frequency of accident (Bullough et al., 2013). Segments with low lighting levels have a greater occurrence of nighttime pedestrian collisions, as stated by Zhou and Hsu in 2007. According to a study by Wanvik (2009), road lighting reduces injury accidents during darkness by an average of 30%. The greatest reduction in accident is shown on motorways, followed by split highways and single carriageway roads, as found by Frith and Jakkett (2015).

Land use configurations impact road safety, particularly pedestrian-involved accidents (Samerei et al., 2021). A major part of pedestrian accident frequency variations is caused by the interplay of density and mixed land use (Ding et al., 2018). The study conducted by Ukkusuri et al. (2012) found that land use categories have a greater impact on the

frequency of pedestrian accident compared to other road users. The study conducted by Kim et al. (2010) found a positive correlation between accident rates and the presence of business and commercial land uses. The presence of retail and commercial establishments in standard strip commercial or large-scale shopping malls has been shown to be strongly correlated with significant growths in the overall number of injurious and fatal accidents. However, when these establishments are situated in smaller, pedestrian-friendly layouts which directly adjoin the street, they have been seen to minimise the likelihood of accidents occurring (Dumbaugh and Li, 2011; Dumbaugh and Rae, 2009). In addition, regions with greater densities of hospitals or schools likely to have a higher incidence of traffic accidents (Wang et al., 2019c). Various land use categories exhibit distinct levels of impact on the frequency of accidents, which can be observed at both the local and neighborhood scales (Yang et al., 2021). The land-use features have an indirect impact on the travel mode choices of persons, which in turn affects accident rates (Lee, 2006). The study conducted by Amoh-Gyimah et al. (2016) found that there is a positive correlation between mixed land use and pedestrian accidents.

Population density increase leads to more pedestrians and cyclists on the streets. Vehicle encounters with pedestrians, cyclists, and drivers may increase, compromising road safety (Soltani and Askari, 2017). In densely populated areas, pedestrian infrastructure, safe crosswalks, and designated bike lanes may prevent accidents. Traffic congestion and limited parking in densely populated regions may stress drivers (Zulfiqar et al., 2023). Stress may lead to lane changes and tailgating, increasing accident risk. Congested roadways may lead to traffic flow restrictions, lane changes, and sudden pauses, increasing the risk of accidents. Houses, offices, and businesses are often located close together in densely populated zones, which increases traffic issues. The complex interplay between population density and pedestrian exposure shows the many connections that affect pedestrian accident rates (Kaplan and Giacomo Prato, 2015). Moreover, the occurrence of fatal and injury accidents has primarily been noted in residential areas and near public establishments that are characterized by relatively low to moderate population densities, particularly on highways with elevated speed restrictions (Rahman et al., 2020).

Commute-related incidents have been linked to factors such as heightened job stress, minor workplace accidents, and occupations with increased noise levels (Bowen and Smith, 2020). According to Schefer and Rietveld (1997), when there is higher density, it can lead to more congestion, slower speeds, and potentially fewer fatal accidents.

Higher job density indicates more commuters to the neighborhood. Congestion and rush-hour traffic may increase the risk of accidents. Commuters entering and exiting the neighborhood may create traffic and driving issues. High-employment density regions may include commercial, residential, and industrial land use. This may lead to commercial vehicles, delivery trucks, and personal cars sharing the road. Such interactions may increase the risk of accidents, particularly around intersections and places with unpredictable traffic flows (Kang, 2023). In employment-dense communities, businesses and commercial activities require parking and loading zones. Parking and loading zones may slow traffic, lead to double parking, and increase accident risk. In densely populated areas, well-designed traffic infrastructure, including signs, road lights, and pedestrian crossings, is vital. Road infrastructure should emphasize safety and suit the needs of road users. Crosswalks, pedestrian-friendly lanes, and cycling lanes may minimise accidents for pedestrians and cyclists (Wu et al., 2021).

Road safety assessments emphasize the role of intersection design. Roundabouts, leg counts, and intersection configurations affect accident rates (Dumbaugh et al., 2013). The nuanced intersection configurations, including three-leg crossroads, are correlated with accident rates for pedestrians and pedal cyclists (Mansfield et al., 2018; Bagloee and Asadi, 2016; Dumbaugh et al., 2013).

Several studies have examined the occurrence rates of accidents in Melbourne, Victoria. A study conducted by Khalifa et al. (2022) revealed

a progressive decrease in the frequency of vehicle collisions at crossings in Victoria between 2018 and 2020. Notably, the year 2018 recorded the greatest incidence of accidents at intersections. A further investigation examined the frequencies of accident involving trams along certain sections of tram routes in Melbourne. This study revealed various characteristics that have an impact on the frequency of these accident, such as the frequency of tram services, the spacing between tram stops, and the overall volume of traffic (Naznin et al., 2016). Moreover, research conducted on vehicle-pedestrian accidents in Melbourne revealed that accidents happening between 7:00 p.m. to 6:00 a.m. exhibit greater severity, with most of these incidents concentrated in the vicinity of hotels, clubs, and bars (Toran Pour et al., 2017). There has been no significant change in the number of severe injuries caused by traffic accidents for many road users in Victoria (Grzebieta et al., 2017). Moreover, prior studies have predominantly focused on investigating driving behavior (McDonald et al., 2020), the impacts of weather condition (Gaffney and Hovenden, 2022), estimating accident severity (Zhu, 2021; Vajari et al., 2020; Nasri et al., 2022), analysing accident types (Zhu, 2020), and predicting accident frequency (Buckis et al., 2015) in the specific geographical context of Victoria, Australia.

This study investigates the spatial and temporal characteristics of accidents in the Greater Melbourne Area (GMA) over a 15-year period using a macro-level analysis. The analysis focuses on the frequency of

accidents per census block to identify areas with a higher risk of accidents. The goal is to propose mitigation policies to reduce accidents in these areas. It also examines how a combination of socioeconomic variables, such as gender and age, and built environment features, including land use types, road and cycling network, distance to the central business district (CBD), and employment density, influence the frequency. The research questions are as follows:

- What spatial patterns manifest in the frequency of traffic accidents, considering changes in temporal distribution, within the GMA?
- Considering the heterogeneity in the spatial distribution of accident occurrences, what factors exert the substantial influence on accident frequency within the confines of the GMA?

This study investigates accident patterns in several geographic regions, with the goal of identifying disparities and commonalities in accident incidence. Comprehending the correlation between geography and accident rates is vital for policymakers when distributing resources for road safety measures. This study is the first to do a comprehensive examination of accident frequencies in connection to the built environment, providing a distinctive viewpoint. Through the analysis of historical data from different locations and time periods, it provides a deeper understanding of the differences in causes that cause accidents,

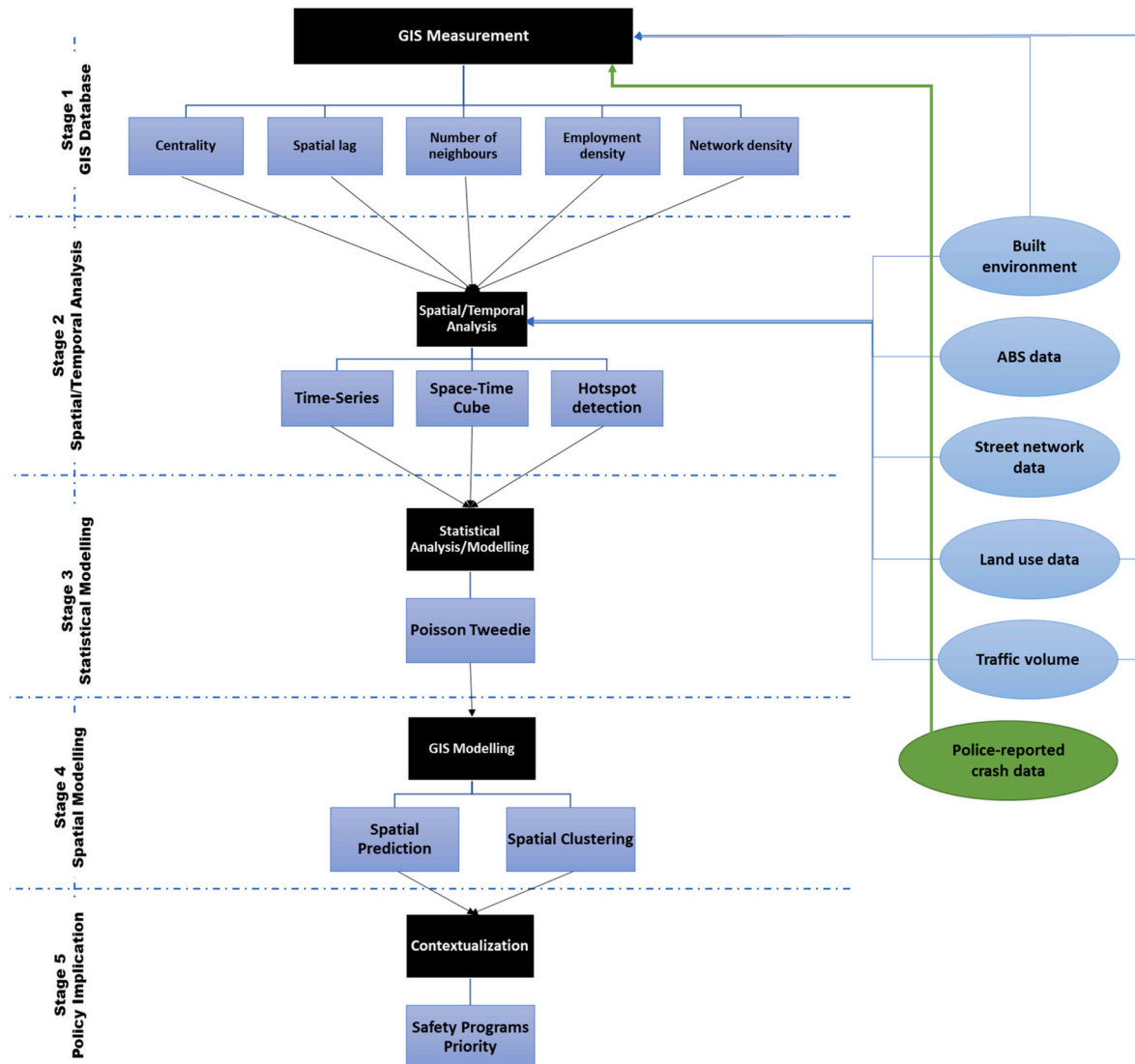


Fig. 1. The research process flowchart.

so providing valuable information for more thorough road safety planning.

The research was conducted via a series of five distinct stages. These stages included several aspects of the study, namely: Stage 1, which involved the creation and management of a Geographic Information System (GIS) database; Stage 2, which focused on the analysis of spatial/temporal patterns; Stage 3, which entailed the use of statistical models; Stage 4, which involved the development of spatial models; and finally, Stage 5, which explored the policy implications derived from the study's findings. Additional information on each stage, including the specific technique used, may be seen in Fig. 1. The modelling and analysis stages were supported by an integrated database comprising police-reported accident data spanning from 2006 to 2020. Additionally, secondary data from government open-source databases including Australian Urban Research Infrastructure Network (AURIN), Victorian Government Dataset and Australian Bureau of Statistics (ABS) were utilized to incorporate socioeconomic, street network, land use, traffic volume, and built environment factors into the analysis.

2. Data and method

2.1. Data

Because of its significance in Australia, GMA has been selected as the case region for this study. Melbourne is the second-largest metropolitan region in Australia in terms of population, and it is one of the most significant economic hubs in the nation. Melbourne, on the other hand, has a low proportion of public transport users (less than 20% for commuting to work), the highest road space per capita of any Australian city and a strong dependence on personal vehicles for work commutes (Redman et al., 2013). This has caused traffic congestion, extended commute times for many commuters, and the risk of accidents. The unit of analysis in this study is Statistical Area Level 2 (SA2), providing for a more nuanced understanding of changes in accident patterns at the suburban and Local Government Area (LGA) levels across time. According to the ABS, SA2 - is intended to reflect a community and ranges in size from around 3000 to 25,000 residents. An LGA is a geographic area that is governed by a local government, which is responsible for providing a range of services to residents within its boundaries, including but not limited to, waste management, parks and recreation, local roads, and community services. There are currently 537 LGAs in Australia, which vary in size, population, and governance structure. GMA includes 34 LGAs. This approach is like some previous research such as Kim et al. (2012), Koloushani et al. (2021) and Toran Pour et al. (2017). The descriptive statistics of variables are shown in Table 1.

2.1.1. Spatial patterns

Examining the spatial patterns of traffic collisions holds significance for multiple reasons. Firstly, it aids in recognizing risk patterns and comprehending the primary components that contribute to the seriousness of accidents on road segments (Mou et al., 2022). Through the examination of the spatial and temporal patterns of accidents, it is feasible to pinpoint places with a higher likelihood of accidents, sometimes referred to as hotspots (Soltani and Askari, 2017; Soltani and Askari, 2014). Subsequently, this data can be utilized to allocate resources in a prioritized manner and execute focused actions aimed at enhancing road safety (Hazaymeh et al., 2022). Moreover, the examination of spatial patterns might unveil the impact of several spatial elements, such as road attributes and land use, on the frequency of accidents (Harirforoush et al., 2019).

While the overall trend in accident frequency in GMA is decreasing, this trend is not consistent across all localities. This suggests that factors such as population growth, changes in infrastructure, and demographic shifts can have a role in determining the incidence of accidents in different parts of the region. One notable trend is the increase in accident frequency in both the outer and inner suburbs. This could be due to

Table 1
Descriptive statistics of variables.

Variables	Definition	Minimum	Maximum	Mean	Std. Deviation
<i>CountAccident</i>	The total count of accidents within the block	140	2766	964	533
<i>P25_29</i>	People in the age range of 25–29	0	5918	1400	916
<i>P60_64</i>	People in the age range of 60–64	0	1790	834	358
<i>EmplDensity</i>	Number of employed people per hectare	0	99	5	10
<i>PopRank</i>	The population rank of the block	0	304	141	90
<i>AvgTrfVol</i>	Average of traffic volume along the street network of the block (aggregated from traffic volume of individual segment). For each street segment, the traffic volume is the sum of traffic traveling in both directions on a two-way road passing a roadside observation point over the period of a full year divided by the number of days in the year.	4760	79,462	25,673	9223
<i>BikLanLngh</i>	Total length of bike routes within the block	2536	81,946	19,724	903
<i>NoRoads</i>	Total number of road segments within the block	3	182	46	29
<i>CntLight</i>	Number of light poles along the street network of the block	1	47	14	10
<i>SpatialLag</i>	Spatial lag of the block: (a spatial lag of a specified variable is computed by taking the weighted average of neighboring polygons, as specified in	1	16	1	1

(continued on next page)

Table 1 (continued)

Variables	Definition	Minimum	Maximum	Mean	Std. Deviation
	the weights matrix. For example: a census tract with three neighboring tracts that have 10, 15, and 20 accident frequency would have a spatial lag of 15. The row-standardized spatial weights matrix is used in the calculation of the spatial lag.)				
Centrality	Centrality of the block	0	64,756	20,485	9223
DistCBD.Ln	Distance to CBD, Ln	0	11	9	2
Parkland	Total number of lots of parkland	0	253	13	118
Commercial	Total number of lots of commercial activity	0	2673	41	167

several factors, such as an increase in traffic volume or changes in driver behavior. It may also reflect transformations within the built environment, such as the introduction of recent road layouts or the expansion of residential or commercial areas. Interestingly, the middle suburbs have remained relatively stable in terms of accident frequency, suggesting that the factors driving the increase in other areas may not be as significant in these parts of the metropolitan region.

2.1.2. Yearly patterns

An analysis of temporal accident event patterns enables the identification of factors and characteristics influencing the number of accidents (Cheng et al., 2021). Analyzing the yearly, monthly and daily traffic accident fatalities can help traffic safety offices determine the optimal timing for safety and enforcement campaigns during periods with high accident rates (Liu et al., 2005). In addition, examining the temporal patterns of accident risk can offer valuable information on the connection between driving schedules and the likelihood of accidents. This data can then be utilized to create more precise accident risk models (Jovanis et al., 1991).

The cumulative number of road accidents recorded annually between the years 2006 and 2020 indicates variations seen over this time frame (Fig. 2). Since its inception in 2006, the cumulative number of accidents was first reported as 45,738. Subsequently, there was a progressive rise in accidents until 2008, with the highest recorded count of 48,724 accidents occurring in that year. From the subsequent years up to 2016, there was a rather stable trend in the occurrence of accidents, characterized by minor variations but no significant departures from the average. Nevertheless, it is worth noting that in the year 2017, a considerable fall in the frequency of accidents was seen, with a recorded figure of 40,934. This downward trend persisted in 2019, when the total number of accidents further declined to 38,178.

The data shows a notable change in 2018, with the number of accidents rising to 41,930. This is a deviation from the declining pattern recorded in the two previous years. It is worth noting that there has been a substantial decrease in road accidents in the year 2020, with the number of events down to 19,510. The significant decline in road activity may be ascribed to the enforcement of lockdown measures in reaction to the COVID-19 pandemic. The temporal trend of accident occurrence for LGAs is displayed on Appendix A.

For most of LGAs, the temporal trend shows a significant decrease in the last period (2018–2021). This can be attributed to COVID-19 pandemic lockdown. Although for some LGAs such as Murrindindi, Port Philip, Yarra Ranges and Melbourne the gaps between the last period and the previous period are more evident. On the other hand, some LGAs experienced a very slight decrease such as Mitchell, Banyule, and Manningham. Interestingly a few LGAs including Moorabool, Darebin, and Hume had higher rates of accident despite the implementation of COVID-19 lockdown.

2.1.3. Monthly and daily patterns

The daily and monthly fluctuations in the incidence of accident by LGA are interesting. Fridays and Thursdays are the least safe days for driving, while March and May are the least safe months. On the other hand, Sundays and January and September are the safest days and months for driving (Appendix B and Appendix C). This figure also show which LGAs are safest ones for most of days, and months, and which ones are less safe during for most of the week, and most of the year. Some LGAs, including Greater Dandenong, Hosbons Bay, Melbourne, Mitchell, Port Philip, Stonnington, and Wyndham, have suffered at least three hazardous days every week; nevertheless, these suburbs are especially secure on weekends. On the other hand, a few LGAs, notably Macedon Ranges, Mitchell, Murrindindi, Nillumbik, and Yarra Ranges, have been noted to be risky on weekends. Intriguingly, the fact that these areas are popular places for amusement and leisure implies that there may be special reasons contributing to their higher incidence of accident. It may be beneficial to study the precise sorts of accident that are happening in these locations and to uncover any common variables or patterns.

Despite daily differences amongst LGAs, their monthly trends are almost consistent. The regularity in monthly trends of accident among LGAs shows that there are common causes for these swings. Weather

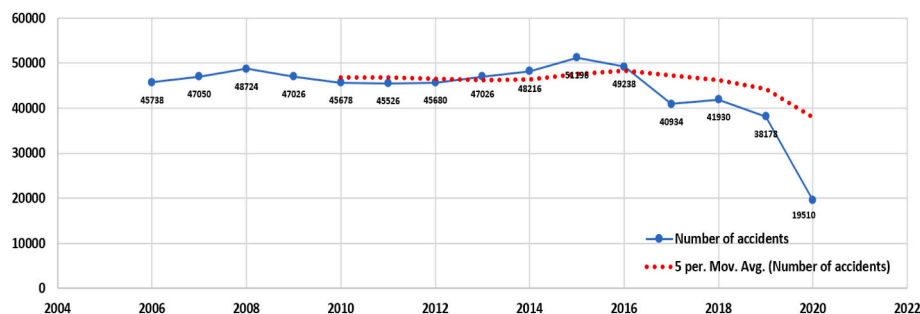


Fig. 2. Yearly fluctuations in accident counts.

conditions, human driving errors, and the quality of the road infrastructure are all possible variables that might affect the monthly patterns of accident. During the winter months (June, July and August), wet weather or reduced visibility due to fog or mist may increase the risk of accident despite the fact the weather condition could be very different across the locations at the same time. Similarly, increasing traffic volume at peak seasons or special events may contribute to increased congestion and a higher accident risk.

2.1.4. The impact of COVID-19

Researchers have examined the effects of COVID-19 on road accidents in different areas. Research conducted in China has demonstrated that there was an initial decrease in traffic accidents during the enforcement of traffic limits, followed by a subsequent increase once the restrictions were withdrawn (Yuan et al., 2023). A study conducted in Chhattisgarh, India revealed a substantial reduction in the number of injuries caused by road traffic accidents during the lockdown period (Sahana et al., 2022). Furthermore, there was a drop in the severity of injuries and the number of fatalities associated with these accidents (Zhang et al., 2022). The impact of COVID-19 was observed to lead to a decrease in the number of crashes and injuries, as well as a decline in road safety performance, but a significant increase in severe accidents (Shaik and Ahmed, 2022).

In GMA, the COVID-19 was associated with lowering the number of accidents. Because of the lockdown, fewer people were on the roads, more people worked from home, and many non-essential companies were closed, resulting in decreased traffic volume and, therefore, a drop in the frequency of accident (Beck and Hensher, 2020). The lockdown also changed driving behavior since individuals were only going for essential activities and were more careful owing to the increased knowledge of the pandemic (Vingilis et al., 2020). More law enforcement and heavier fines for traffic offences came from the lockdown measures, which may have discouraged drivers from participating in unsafe driving practices (Martin, 2022).

Melbourne underwent various restrictions and was labelled as the most restricted city in Australia, with stay-at-home orders lasting a total of 267 days (Kelly et al., 2021) (Fig. 3). Melbourne saw a decline in the number of accidents happening (decreasing by 35% from 1384 accident in April 2019 to 901 in April 2020). On May 2020, 28% less people involved in accident than in May 2019, but in the next month, when Melbourne's lockdown was the tightest, there was a big drop in the number of accidents. Between the June 2019 and June 2020, the number of accidents went down by 49%. This decreasing trend continued until October, when the difference between October 2020 and October 2019 became 93%. In average GMA experienced 63% reduction in accident frequency within seven months due to COVID-19 lockdown.

2.2. Analytical methods

2.2.1. Space-time cube analysis

The integration of spatial and temporal dimensions in GIS would enable the examination of dynamic patterns via the use of space-time cube analysis. A three-dimensional "Space-time Cube" is generated, with the x and y axes representing spatial dimensions, while the z axis represents the temporal dimension. This technique reveals the temporal and spatial variations in characteristics, therefore elucidating patterns and correlations that may go unnoticed in two-dimensional cartographic representations (Wu et al., 2022; Yoon and Lee, 2021; Sen et al., 2022).

Space-time cube analysis entails utilizing wavelet decomposition to break down event data to enhance the efficiency of solving sparse matrix problems (Surodjo et al., 2023). Through the space-time cube analysis of accidents, it is feasible to pinpoint locations with high accident rates and get insight into the underlying reasons (Bodolo, 2023). Furthermore, the implementation of a traffic accident time-space analysis can facilitate the gathering and examination of accident data, pinpoint regions with a high incidence of accidents, and suggest preventive actions (Zhang and Shi, 2019).

The data structure generated may be conceptualised as a tridimensional cube composed of spatial and temporal bins, where the x and y dimensions correspond to spatial dimensions, and the t dimension represents time (Cheng et al., 2018). Each bin is assigned a specific location in both space (x, y) and time (t). Bins that have the same (x, y) coordinates are assigned a common location identifier. Bins that have the same duration are assigned the same time-step ID. By using spatial positioning techniques, a bin time series may be generated, resulting in a longitudinal column known as a bin time sequence. This bin time sequence comprises a series of unit cubes arranged vertically.

2.2.2. The Tweedie modelling

Tweedie modelling is employed for the prediction of traffic accident counts (Debrabant et al., 2018; Saha et al., 2020). The model effectively addresses the issues of zero-inflation and overdispersion, rendering it well-suited for the analysis of accident data. Researchers have employed this method to identify persistent dark spots in Funen, Denmark. They propose that these locations be subjected to additional scrutiny and highlight the possibility of future planning for accident prevention in these areas (Meganfi et al., 2023).

In this study, the Poisson-Tweedie model was used due to its flexibility in managing both overdispersion and under-dispersion of the accident data (Debrabant et al., 2018). The Tweedie class of distribution expands upon the Poisson distribution by including supplementary parameters to accommodate for the phenomenon of overdispersion. The Tweedie distribution may be seen as an extension of the Compound Poisson distribution, whereas the Poisson-Tweedie model integrates components from both distributions. The preference to use this model in accident analysis has grown, because compared to the conventional

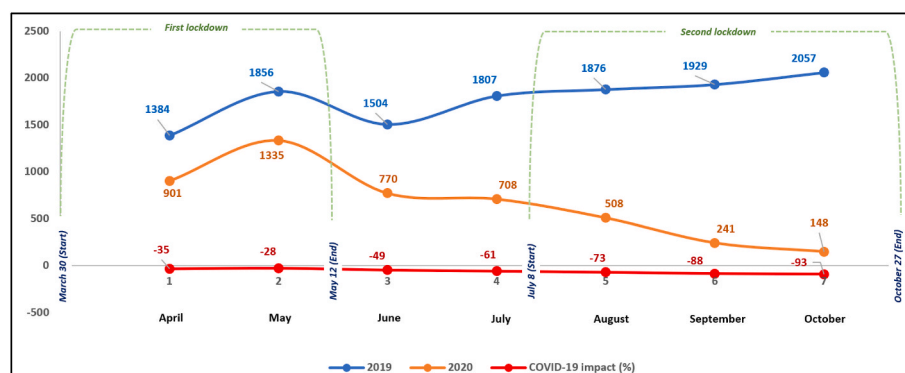


Fig. 3. The impact COVID-19 on accident frequency.

Poisson distribution, it offers a more accurate representation of the data distribution and a method for better capturing the inherent variability of count data (Saha et al., 2020; Goyani et al., 2022).

The Tweedie distribution is suitable for variables that are expressed as a combination of Poisson mixtures of Gamma distributions. This distribution is considered “mixed” because it incorporates characteristics of both continuous distributions (which take non-negative real values) and discrete distributions (which have a positive probability mass at a single value, 0). The dependent variable should possess numerical characteristics, with data values that are equal to or higher than zero. The mathematical explanation of this model is given below (Bonat et al., 2018). The Poisson-Tweedie distributions can be understood as a mixture of Poisson and Tweedie distributions. Consequently, we begin by considering an exponential dispersion model, which can be represented as:

$$f(z; u, \varphi, p) = a(z, \varphi, p) \exp\{z \varphi - k_p(\psi / \varphi)\} \quad (1)$$

In this formula, μ represents the mean, which is given by $E(Z) = k'_p(\psi)$, the dispersion parameter is denoted by φ , where $\varphi > 0$. The canonical parameter is represented by ψ and $k_p(\psi)$ refers to the cumulant function. The variance, denoted as $\text{Var}(Z)$ may be expressed as $\text{Var}(Z) = \psi V(\mu)$, where $V(\mu)$ is referred to as the variance function and is defined as $k''_p(\psi) V(\mu)$. Tweedie densities exhibit power variance functions denoted as $V(\mu) = \mu^p$, where $p \in (-\infty, 0] \cup [1, \infty)$ is the index that defines the distribution. The Tweedie random variable Z is denoted as $Z \sim \text{Tw}_p(\mu, \varphi)$. The support of this distribution is contingent upon the magnitude of the power parameter. For any value of p greater than or equal to 2, the support corresponds to the set of positive values. For values of p between 1 and 2, the support corresponds to the set of non-negative values. Finally, when p equals 0, the support corresponds to the set of real values. In the instances, the symbol μ belongs to the set Ω , where Ω represents the convex support. Specifically, Ω denotes the interior of the closed convex hull of the distribution support associated with the aforementioned occurrences. In the case when $p < 0$, the support aligns with the set of real values once again, but the expectation μ assumes a positive value.

The Poisson-Tweedie family is defined by the accompanying hierarchical condition:

$$Y|Z \sim \text{Poisson}(Z) \quad (2)$$

$$Z \sim \text{Tw}_p(\mu, \varphi) \quad (3)$$

In order to guarantee that Z is a non-negative value, it is necessary to impose the condition $p \geq 1$. The Poisson-Tweedie model, as described by Jørgensen and Kokonendji (2016), may be characterized as an overdispersed factorial dispersion model. The probability mass function corresponding to values of $p > 1$ may be expressed as follows:

$$f(y; \mu, \varphi, p) = \int_0^\infty \frac{z^y \exp - z}{y!} a(z, \varphi, p) \exp\{z\psi - k_p(\psi)\} / \varphi \, dz \quad (4)$$

The integral denoted as (4) does not possess a straightforward analytical expression, except for the specific scenario when it corresponds to the negative binomial (NB) distribution. This occurs when the parameter p is equal to 2, resulting in a Poisson gamma mixture. When the value of p is equal to 1, the integral denoted by (5) is substituted with a summation, resulting in the NTA (Neyman Type A) distribution. The technique of the estimating function uses the quasi-score and Pearson estimating functions to estimate regression and dispersion parameters, respectively. The quasi-score function for the parameter β is expressed in the following form:

$$\psi_\beta(\beta, \lambda) = \left(\sum_{i=1}^n \frac{\partial \mu_i}{\partial \beta_1} C_i^{-1} (Y_i - \mu_i), \dots, \sum_{i=1}^n \frac{\partial \mu_i}{\partial \beta_q} C_i^{-1} (Y_i - \mu_i) \right)^T \quad (5)$$

Where $\frac{\partial \mu_i}{\partial \beta_j} = \mu_i x_{ij}$ for $j = 1, \dots, q$. The entry (j, k) of the $q \times q$ sensitivity matrix for ψ_β has a form of:

$$S_{\beta_{jk}} = E \left(\frac{\partial}{\partial \beta_k} \psi_{\beta_j}(\beta, \lambda) \right) = - \sum_{i=1}^n \mu_i x_{ij} C_i^{-1} x_{ik} \mu_i \quad (6)$$

Similarly, the entry (j, k) of the $q \times q$ variability matrix for ψ_β is given by

$$V_{\beta_{jk}} = \text{Cov}(\psi_{\beta_j}(\beta, \lambda), \psi_{\beta_k}(\beta, \lambda)) = \sum_{i=1}^n \mu_i x_{ij} C_i^{-1} x_{ik} \mu_i \quad (7)$$

The form of the Pearson estimating function for the dispersion parameters is as follows:

$$\psi_\lambda(\lambda, \beta) = \left(- \sum_{i=1}^n \frac{\partial C_i^{-1}}{\partial \lambda} [(Y_i - \mu_i)^2 - C_i], - \sum_{i=1}^n \frac{\partial C_i^{-1}}{\partial \rho} [(Y_i - \mu_i)^2 - C_i] \right)^T \quad (8)$$

The Pearson estimating functions are a class of unbiased estimating functions for the parameter λ , which are based on the squared residuals $(Y_i - \mu_i)^2$ and include the expected value C_i . The implementation of the estimate for the Poisson-Tweedie model in R is facilitated by the *mclgm* Library (Bonat, 2016). This study employs the Tweedie distribution with the *Identity* link function, as specified in the methodology.

2.2.3. Clustering

To better understand the complex patterns and interrelationships present in the spatially referenced factors impacting accident frequency, multivariate spatial clustering was employed to group the relevant variables derived from the Tweedie test (Ohana-Levi et al., 2019). Since the chosen variables use different measurement units or scales, a standardisation procedure was performed as a pre-processing step to ensure consistency across analyses. To further include the spatial dimension into the clustering process, spatial weight matrices were defined to account for the proximity, adjacency, and other spatial associations between individual data points. Subsequently, the k-means clustering technique can be utilized to examine the geographical distribution pattern of accidents, partitioning the traffic accident area into several categories (Wu et al., 2021).

Based on the F-test curve, the default K-means clustering technique was selected, and the optimal number of clusters was 10. Using the normalised data and geographical linkages, the K-means method creates a network of clusters, each of which is made up of geospatial entities that are similar in both attribute values and spatial context. The resulting clusters are multifaceted items that reveal previously unseen aspects of geographical data.

The resulting cluster map helped us have an interpretive discussion and get useful insights about the safety policy priorities in GMA, allowing us to propose and implement LGA-level safety measures. The evaluation of clustering performance is quantified by using the Calinski-Harabasz pseudo-F-statistic, which represents the ratio between the variation seen between clusters and the variance observed inside the clusters. In other terms, the F concept refers to a ratio that signifies the level of resemblance within a group and the level of diversity between different groups (Hinde et al., 2007):

$$F = \frac{\frac{R^2}{n_c - 1}}{\frac{1 - R^2}{n - n_c}} \quad (9)$$

where:

$$R^2 = \frac{SST - SSE}{SST} \quad (10)$$

SST represents the between-cluster differences while SSE represents the within-cluster similarity:

$$SST = \sum_{i=1}^{n_c} \sum_{j=1}^{n_t} \sum_{k=1}^{n_v} (V_{ij}^k - V^k)^2 \quad (11)$$

$$SSE = \sum_{i=1}^{n_c} \sum_{j=1}^{n_t} \sum_{k=1}^{n_v} (V_{ij}^k - V_t^k)^2 \quad (12)$$

n = the number of features

n_i = the number of features in cluster i

n_c the number of classes (clusters)

n_v = the number of variables used to cluster features

V_{ij}^k = the value of the k th variable of the j th feature in cluster i

V^k = the mean value of the k th variable

V_t^k = the mean value of the k th variable in cluster i

The K-means method was used in this study. The objective of the K-means method is to divide characteristics into clusters in such a way that the dissimilarities among the features inside each cluster, across all clusters, are minimised. The K-means method starts by determining initial centroids, which serve as seeds for the subsequent growth of each cluster. As a result, the quantity of seeds will consistently correspond to the number of clusters. The first seed is chosen in a random manner.

The process of selecting the remaining seeds include a random element, but it also incorporates a weighting mechanism that prioritises the selection of seeds that are located farther away in data space from the current collection of seed characteristics. After the identification of the seeds, all features are allocated to the seed feature that is closest in the data space. A mean data centre is calculated for each cluster of features, and afterwards, each feature is transferred to the centre that is closest to it. According to Jain (2009), the iterative procedure of calculating the mean data centre for each cluster and thereafter reassigning features to the nearest centre continues until the membership of clusters reaches a state of stability, with a maximum limit of 100 iterations.

2.2.4. Prediction by GWPR

Geographically Weighted Poisson Regression (GWPR) is designed to model count data, in which the dependent variable adheres to a Poisson distribution. This methodology expands upon the framework of Geographically Weighted Regression (GWR) to effectively model count-based data, namely accident frequency (Soltani et al., 2021). The utilization of GWR has a lengthy historical background in accident research, as noted by Rosenlieb et al. (2018). The study conducted by Gomes et al. (2017) found that GWPR and Geographically Weighted Negative Binomial Regression (GWNBR) outperformed the Generalized Linear Model (GLM) in accident prediction models. Additionally, these methods were more effective in capturing the geographical variations in accident frequency. The application of GWPR enhanced the accuracy of accident analysis by accounting for the non-stationary correlations between accident and variables (Shariat-Mohaymany et al., 2015).

The GWPR model to predict the frequency of accidents basing on count data, while also considering the geographical variability of the measured variables. The GWPR technique involves the estimation of individual Poisson regression models for each site within a defined bandwidth. The estimate procedure involves the computation of the local rate parameter and Poisson likelihood for every site. In this work, the *GWmodel* Library in R was used with the following assumptions (Brunsdon et al., 2002).

- Kernel function: bi-square;
- Adaptive bandwidth: 150 (number of closest neighbors);
- Regression points: the same positions as observations are applied, and
- Distance metric: A distance matrix is given for calibrating the model.

Further elaboration on the structure of this model is provided below. A Conventional Poisson Regression Model:

$$y_i \sim \text{Poisson} \left[N \exp \left(\beta_0 + \sum_k \beta_k x_k \right) \right] \quad (13)$$

where,

β_0 = intercept

β_k = coefficient of independent variable x_k

N = offset variable

Geographically Weighted Poisson Regression (GWPR)

$$y_i \sim \text{Poisson} \left[N_i \exp \left(\beta_0(u_i, v_i) + \sum_k \beta_k(u_i, v_i) x_{k,i} \right) \right] \quad (14)$$

where,

$\beta_0(u_i, v_i)$ = intercept parameter specific to location i

$\beta_k(u_i, v_i)$ = coefficient of independent variable x_k at location i and

N_i = offset variable at i th location

$(u_i, v_i) = x \sim y$ coordinate at i th location

3. Findings

3.1. Space-time cube analysis

The Space-time Cube of accidents for Melbourne, provides valuable insights into the spatial/temporal patterns of accidents over the period of 2016–2021. The hotspots and cold spots of accidents as well as the transformation of these spatial clusters provides insights into locations with higher risk factors or areas where safety measures are effective. The accident patterns in inner city areas and outer city areas are different. According to the Space-time Cube, accidents are more prevalent in densely populated urban centers and along the major arterials connecting the central areas to less populated suburban regions. The changes of hotspots over time shows that certain areas in GMA are consistently problematic, although the patterns have shifted from year to year. Furthermore, in specific time periods, specific locations become high-risk areas for accidents. This could be due to events like festivals, construction, or changes in road infrastructure. The process involves the enumeration of points inside individual cube bins, the computation of summary field statistics, and the assessment of temporal patterns in bin values at each specific site via the use of the Mann-Kendall statistic. By combining the obtained polygons on SA2 level, it is concluded that out of 309 statistical blocks, 218 statistical blocks are statistically significant (Level of confidence >90%) in explaining accident frequency in both dimensions of space and time.

The Space-time Cube analysis shows that the neighborhoods in the northern and western LGAs have an upward trend in accidents, with Melton and Wyndham having the strongest upward trend above. On the other hand, the eastern and southern areas of GMA have experienced a downward trend in accident frequency, with Knox and Stonnington having the strongest downward trend. Additionally, the analysis shows that central Melbourne has also experienced a downward trend in the number of accidents. As shown in the previous studies, the City of Melbourne has the highest density of accident frequency (Hovenden and Liu, 2020). These findings are supported by Fig. 4, which provides a visual representation of the trends in different regions of Melbourne. Based on the sign of the Z-value, the statistical regions with a P-value less than 0.1 are identified. The upward and downward trends in these regions can provide insights into the underlying factors driving the trend in accidents, such as changes in traffic patterns, road infrastructure, or driver behavior. Year 2020 is ignored due to COVID 19.

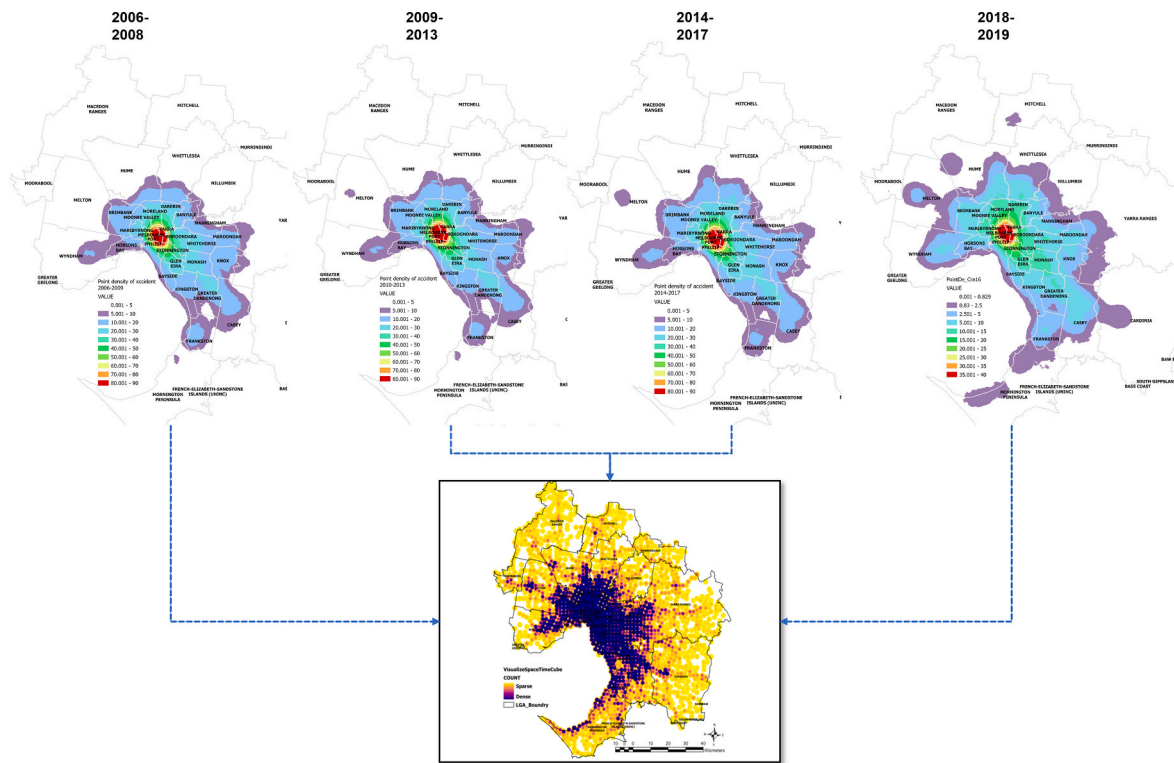


Fig. 4. Visualization space time cube of accidents 2006–2019 (down); kernel density (top).

3.2. Poisson tweedie modelling

Before doing a Tweedie Modelling, we have discovered the existing patterns of the explanatory variables and any potential correlations among them. A total of 14 variables were taken into account in the Tweedie modelling process. These variables include CountAccident, which represents the discrete variable for accident frequency in each SA2 block, as well as 13 covariates as outlined in Table 1. The specifications of the Tweedie model with traffic volume including metrics of model's assessment included: Akaike's Information Criterion (AIC = 19.832); Finite Sample Corrected AIC (AICC = 21.603); Bayesian Information Criterion (BIC = 74.724); and Consistent AIC (CAIC = 89.724).

Moreover, the Omnibus test confirmed the overall significance or fit of the Tweedie model. (Likelihood Ratio Chi-Square = 514.244; p-value < 0.000). Table 2 presents the coefficients of the explanatory variables, together with their corresponding significance levels and Wald values. The table shows that each of the 13 variables has statistical significance at 90% confidence level or over. Overall, each coefficient's sign provides insights into the direction of impact on accident frequency, keeping other factors constant.

The specifications of the Tweedie model without traffic volume including metrics of model's assessment included: Akaike's Information Criterion (AIC = 4084.406); Finite Sample Corrected AIC (AICC = 4085.895); Bayesian Information Criterion (BIC = 4136.118); and Consistent AIC (CAIC = 4150.118).

The power parameter (p) is a key characteristic of the Tweedie distribution. It determines the shape of the distribution and controls the relationship between the mean and the variance of the response variable. A value of p close to 1 indicates a distribution with a high probability of zeros, while a value close to 2 indicates a distribution with a more continuous and symmetric shape. In our case, a power parameter of 1.5 suggests a distribution that is between a Poisson distribution ($p = 1$) and a Gaussian distribution ($p = 2$).

The scale parameter, also known as the dispersion parameter, is a

Table 2

Parameter estimates.

Parameter	B	Std. Error	Hypothesis Test	
			Wald Chi-Square	Sig.
(Intercept)	−28.061***	8.4314	11.108	<0.001
P25_29	0.087***	0.0212	16.806	<0.001
P60_64	0.150***	0.0413	13.155	<0.001
EmplDensity	10.814***	2.4263	19.866	<0.001
PopRank	−0.444***	0.1691	6.888	0.009
AvgTrfVol	0.001*	0.0008	2.738	0.098
BikLanLngh	0.005***	0.0015	11.461	<0.001
NoRoads	8.956***	0.7534	141.304	0.000
CntLight	−11.923***	2.7890	18.276	<0.001
SpatialLag	24.174***	7.7637	9.695	0.002
Centrality	0.005***	0.0014	12.496	<0.001
DistCBD_Ln	22.935***	7.0690	10.527	0.001
Parkland	0.010*	0.0059	2.992	0.084
Commercial	−0.323***	0.1141	7.991	0.005
Dep. Variable	Count of accident per each block		No. Observations	309
Model	GLM		Df Residuals	295
Model Family	Tweedie		Df Model	13
Link Function	log		Scale	4.3445
Method	IRLS		Log-Likelihood	−2216.5
Date	Sun, Oct 29, 2023		Deviance	1174.6
Time	19:57:05		Pearson chi2	1.28e+03
No. Iterations	13		Pseudo R-squ. (CS)	0.7912
Covariance	nonrobust			
Type				
Power	1.5 "			
Parameter				

*p-value<0.10; **p-value<0.05; ***p-value<0.01.

measure of the variability or dispersion of the response variable around the mean. In our output of forecasting the accident frequency in GMA, the scale parameter is specified as 4.345. This value represents the estimated dispersion of the response variable. A higher scale parameter

indicates higher variability in the data. It is important to note that the scale parameter is specific to the Tweedie distribution and may not have a direct interpretation in the same way as the standard deviation in a Gaussian distribution.

A positive coefficient for the population in the age range 25–29 signifies that a rise in the population of this age group is linked to an elevated incidence of accidents. Younger drivers may display more hazardous driving habits. A positive coefficient for the population in the age range 60–65 indicates that a growth in the population of this age group is correlated with a greater incidence of accidents. This phenomenon may be attributed to variables such as possible alterations in driving behaviours or increased knowledge of road safety among elderly drivers.

The positive coefficient for employment density indicates that a larger concentration of employment in a suburb is linked to a greater occurrence of vehicle accidents. This phenomenon may be attributed to increased traffic congestion and interplay between cars in regions with elevated employment levels.

The negative coefficient for the population rank indicates an inverse relationship between population rank (more populated suburbs) and accident frequency, implying that higher population ranks are linked with lower accident frequencies. This might be attributed to enhanced infrastructure, more awareness, or more efficient traffic management in densely populated regions. A positive coefficient for average traffic volume indicates a direct correlation between higher traffic volume and an increased incidence of accidents (Allan et al., 2022). The link between traffic and accidents is straightforward, since an increase in traffic might result in a higher probability of accidents. The positive coefficient for the length of the bike lane indicates that an increase in the extent of bike lane infrastructure is linked to a greater frequency of accidents (Soltani et al., 2022). This might suggest intricate interplays among various categories of road users or particular road configurations.

A positive coefficient for the number of road segments indicates that an increase in the number of road segments is correlated with a greater incidence of accidents. This suggests that a more concentrated road network results in heightened vehicular encounters. A negative coefficient for the number of streetlights indicates that a decrease in the number of streetlights is linked to an increase in the frequency of accidents. This observation suggests that well-illuminated locations promote safer driving or may be susceptible to other variables that contribute to safety.

The positive coefficient for the spatial lag variable indicates that there is a direct relationship between the accident frequencies in adjacent suburbs and the accident frequency in the present suburb. In other words, if the accident frequency is greater in the neighboring suburbs, it is also likely to be higher in the current suburb. This suggests that the frequency of accidents in the present suburb may be influenced by accident trends in nearby places.

A positive correlation for centrality indicates that a greater level of centrality, in relation to other suburbs, is correlated with a higher incidence of accidents. It is important to understand that being centrally located in Melbourne does not necessarily mean being closer to the CBD region. This phenomenon may be attributed to elevated levels of vehicular congestion or heightened levels of activity in core regions. The positive coefficient for distance to the Central Business District (CBD) indicates that increased distance from the CBD is correlated with a greater incidence of accidents. This observation suggests that there may be variations in driving habits or road conditions between suburban districts and the city core.

Expanding the area designated for parks might potentially draw a larger number of walkers and bikers, leading to an elevated likelihood of encounters and collisions. Furthermore, the parklands that have lower speed restrictions may potentially incentivize vehicles to partake in risky driving practises. A negative coefficient for the floor space of commercial land use activities indicates that a bigger business floor area is linked to a decreased probability of accidents. This phenomenon may

be attributed to targeted traffic control measures implemented in business zones or a decrease in aggressive driving habits inside these areas.

3.3. Prediction of accident frequency by GWR

GWR technique enabled us to have a prediction of accident frequency for the next 15 years, considering the spatial heterogeneity. The prediction is based on the diverse impacts of measure variables introduced on Figs. 5 and 6 it is anticipated that downtown Melbourne will continue to be the primary location for accident hotspots, while the inner suburbs are expected to have a decrease in the occurrence of accidents, thereby enhancing safety levels. Certain mountainous regions situated in the eastern part of Melbourne, namely Cardinia and Yarra Ranges, are anticipated to see a heightened incidence of vehicular accidents. Likewise, several western suburbs, like Melton, Hume, Moorabool, and Brimbank, are anticipated to have an increasing incidence of accidents. Interestingly, there is a growing trend of increasing accident frequency in beach towns located either in the western region, such as Wyndham and Greater Geelong, or in the eastern region, including Frankston, Mornington Peninsula, and French-Elizabeth-Sandstone Island.

3.4. LGA safety classification

The clustering analysis aimed to shed light on the distribution of road accident risk scores across various LGAs, providing valuable insights into road safety patterns. Utilizing K-means clustering algorithm, LGAs were grouped into distinct clusters based on their road accident risk scores. However, it is essential to emphasize that the results are derived from the provided risk scores and the algorithm's grouping decisions, necessitating careful interpretation within the context of the study's objectives (Fig. 7).

Within the Cluster 1 (lowest risk score), LGAs like Bass Coast, Baw Baw, French-Elizabeth, Greater Geelong, Murrindindi, and South Gippsland exhibit notably lower road accident risk scores. This cluster implies that these areas may have implemented effective road safety measures or possess favorable geographical and demographic conditions that contribute to their safer road environments. The LGAs in Cluster 2, including Cardinia, Darebin, Frankston, Hobsons Bay, Hume, Melton, Mitchell, Moonee Valley, Moorabool, Moreland, Mornington Peninsula, and Whittlesea, demonstrate a moderate range of road accident risk scores. This cluster suggests that these LGAs may encounter a balanced blend of risk factors, possibly reflecting varying traffic densities and infrastructural conditions.

Cluster 3 comprises Banyule, Bayside, Brimbank, Casey, Glen Eira, Kingston, Knox, Maribyrnong, Maroondah, Yarra, and Yarra Ranges, showcasing a diverse range of risk scores. The clustering of these LGAs implies that they share commonalities contributing to comparable levels of road accident risk, which could encompass factors such as road design, population density, or socio-economic conditions. LGAs with intermediate to higher risk scores, such as Greater Dandenong, Macedon Ranges, Nillumbik, and Wyndham, form Cluster 4. These LGAs may experience specific risk factors that elevate their road accident risk, including factors like commercial activity, road infrastructure complexity, or urban planning.

Lastly, the Cluster 5 (highest risk score), encompasses Boroondara, Melbourne, Port Phillip, Stonnington, and Whitehorse, reflecting towns with elevated road accident risk scores. This cluster might be indicative of densely populated urban centers or areas with intricate road networks, potentially magnifying the likelihood of accident due to increased traffic interactions.

These clusters provide a foundation for further investigation into the nuanced interplay of factors contributing to road accident risk and could serve as a basis for targeted road safety interventions and policies in these diverse townships. The Victorian Road Safety Program achieved over 180 safety enhancements on highways, roads, and streets.

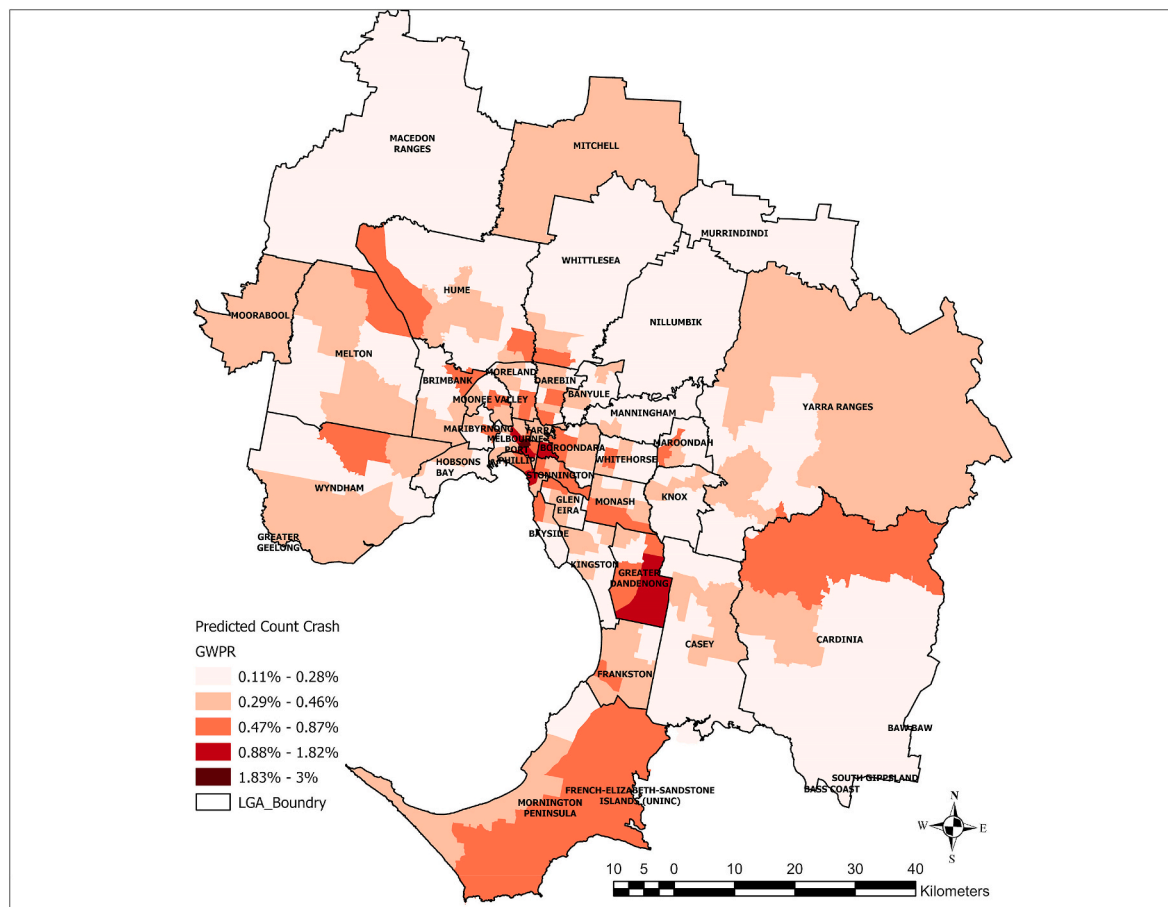


Fig. 5. Prediction count of accident FOR 2035.

Upgrades include shoulder sealing, rumble strips for control, flexible barriers to prevent accident, and improved cyclist-pedestrian safety (Victoria State Government, 2023). In mid-2023, virtual community meetings gathered regional input to enhance program recommendations (Department of Transport and Planning, 2022).

4. Discussion

Melbourne's metropolitan geography, being the core hub of Victoria, exhibits a diverse range of characteristics that significantly influence its transport dynamics. These factors include job density, road infrastructure, demography, and geographic centrality (Li and Dodson, 2020). This paper explored the results of spatial and non-spatial analyze that were undertaken to explain the frequency of accidents in GMA. Furthermore, this research provided an interpretation of the coefficients within the framework of the distinctive characteristics of the metropolitan region as below.

There is a positive correlation between the age range of 25–29 and an increased incidence of accident. Moreover, the coefficient associated with the population aged 60–65 has a positive correlation. This phenomenon may be attributed to alterations in driving behaviours or disparities in road safety consciousness among elderly drivers (Vaa, 2007; Devlin and McGillivray, 2014). Although the exact reasons behind accidents across these two age demographics have not been determined, previous studies indicate unique factors that impact driving safety. The study conducted by Chen and Wang (2019) found that traffic offences and vehicle behavior are important determinants for younger drivers, whereas weather and time of day have a notable impact on older drivers in Canada. It is worth mentioning that young male drivers have a higher likelihood of being involved in accidents, particularly in Madrid, Spain.

They often have difficulties like as losing control on corners and accidents happening in the dark (Clarke et al., 2002). Excessive speed is identified as a significant cause in incidents involving younger drivers (Elliott et al., 2003). Furthermore, research conducted in Norway and Sweden has established a correlation between the engagement of older drivers in accidents and their visual search behavior, attentiveness, and hazard perception (Levin et al., 2009). In addition, older drivers who have lower MMSE (Mini-Mental State Examination) scores, indicating cognitive impairment, may have inferior driving performance (Pyun et al., 2018).

Our analysis found a positive correlation between higher job density inside a LGA and an elevated occurrence of road accidents. This phenomenon may be ascribed to increased traffic congestion and heightened vehicle interactions in regions characterized by vibrant employment activities (Kim and Park, 2018). Multiple studies have emphasized the link between different causes and the frequency of accidents in particular regions. Accident frequency generally rises when the jobs-housing ratio increases (Xu et al., 2017). The correlation between employment and housing has a substantial impact on the occurrence of traffic accidents (Pulugurtha et al., 2013). Moreover, there is a correlation between a rise in employment rates in a neighborhood and temporary increases in injuries (Wier et al., 2009). Furthermore, there are statistically significant correlations between population, job numbers, and accidents, as demonstrated by Kim et al. (2006). Areas with higher local job and population concentrations tend to experience increased accident frequency and mortality rates (Xu et al., 2014).

The frequency of traffic accidents in an urban region is directly influenced by the size of the population, exhibiting a superlinear correlation (Carmen et al., 2020). Conversely, there is a correlation between the size of a municipality and a reduction in the likelihood of road

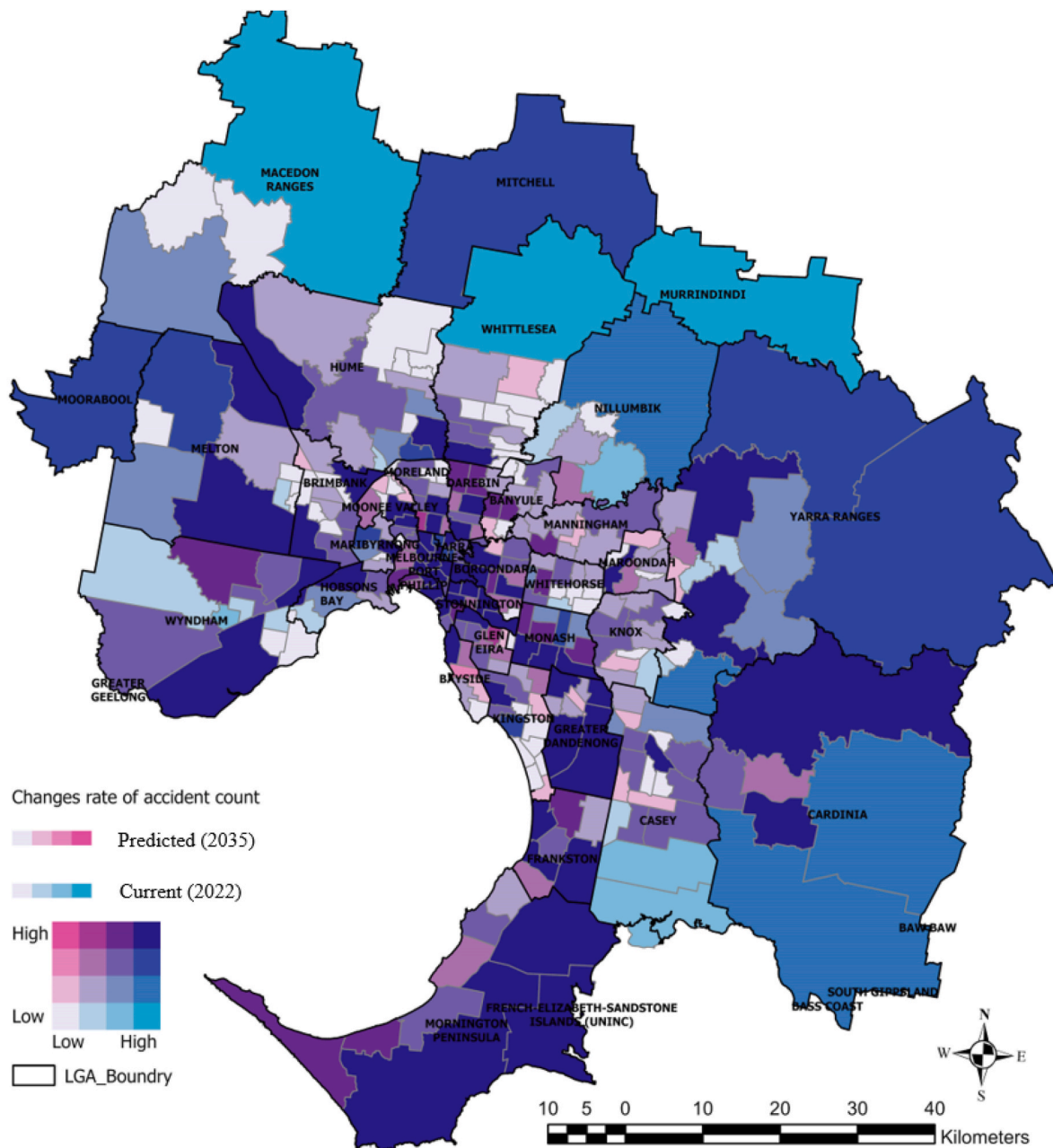


Fig. 6. The current status (2020) versus the predicted year (2035).

accidents, indicating that larger towns tend to experience fewer accidents on the road (Scheiner et al., 2012).

Our analysis confirmed that the traffic volume measured by Annual Average Daily Traffic (AADT) is an important factor in predicting accident frequency, according to Wang et al. (2019b). Vehicle kilometers travelled (VKT) or Vehicle Miles Travelled (VMT) have been associated with a higher occurrence of traffic accident, especially when data on AADT is not available (Rhee et al., 2016; Liu and Sharma, 2017). In similar, a national comparison of crash data from Melbourne, Perth, and urban Queensland examined the severity of accidents linked to various speed ranges (Cameron, 2015).

According to our results, there is a positive correlation between the presence of a substantial bike lane infrastructure and an increased incidence of accident. The discovery implies complicated connections among several categories of road users, perhaps revealing sophisticated manoeuvres or spatial dynamics involving bikes, vehicles, and pedestrians (Kim and Kim, 2015; Vandenbulcke et al., 2014). Segments with

bicycle lanes had a 30% higher likelihood of experiencing accident, according to Kondo et al. (2018). Additionally, they considerably decrease mortality severity in Seoul (Lim et al., 2016). Nevertheless, the implementation of dedicated bicycle lanes has been proven to be the most effective measure in minimising the likelihood of accidents for cyclists (Morrison et al., 2019). The risk level of protected bike lanes changes based on the degree of separation (Cicchino et al., 2020). Remarkably, the existence of cycling facilities on roads in Iowa is associated with a significant reduction in the likelihood of accident by as much as 60% (Hamann and Peek-Asa, 2013). Furthermore, the number of accidents per bicyclist decreases as the frequency of bicycle usage increases (Nordback and Marshall, 2010). The density of primary roadways is positively correlated with all levels of accident severity (Azimian et al., 2021). The density of roads has a favorable impact on the occurrence of traffic accident (Wang et al., 2019a).

The presence of a negative coefficient in relation to the number of streetlights in our model provides evidence that well-lit locations

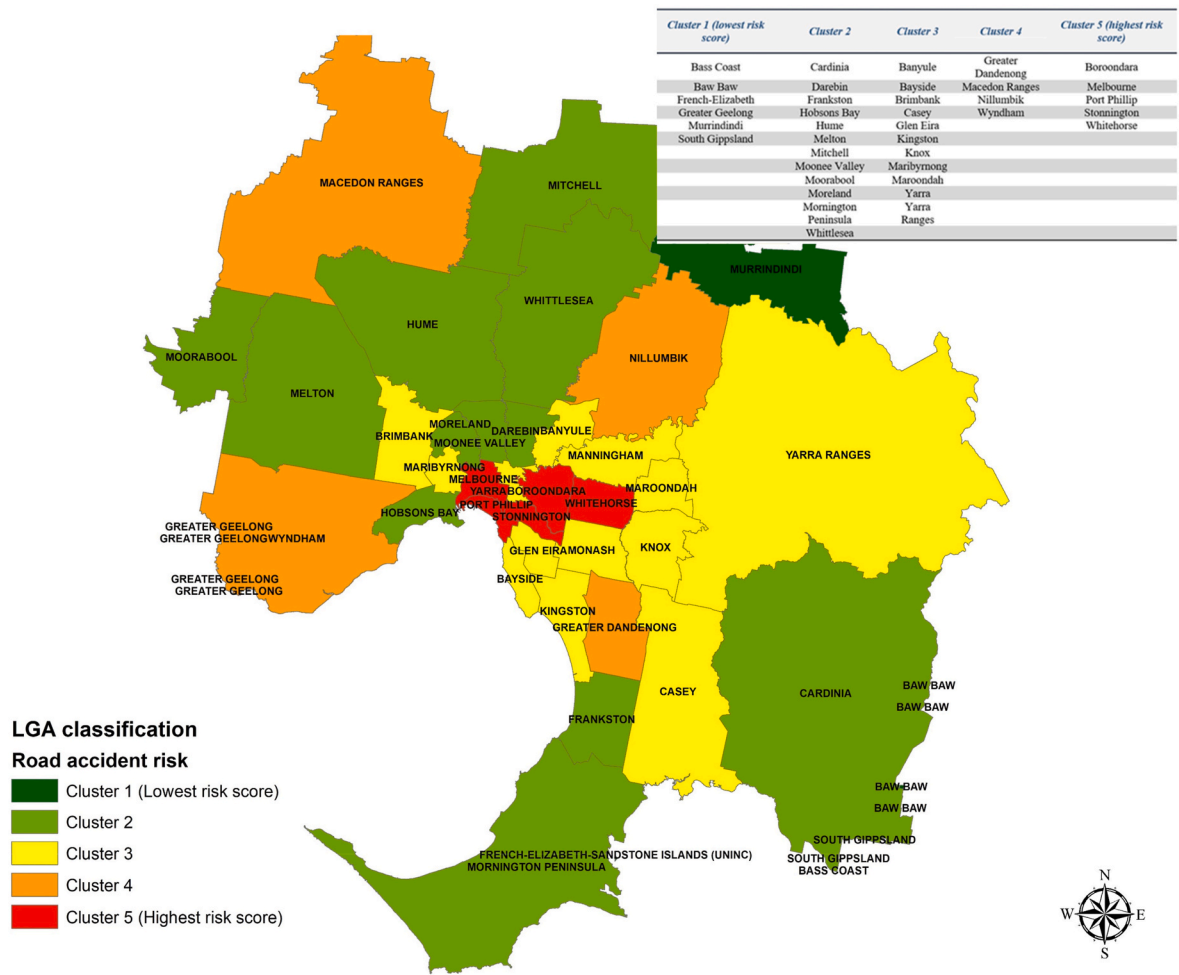


Fig. 7. LGA classification based on road accident risk.

contribute to road safety, although other variables, such as the adoption of safer driving practises, may also have an influence (Reynolds et al., 2009) This highlights the significance of comprehending the interaction of illumination, driver conduct, and other contextual factors within the intricate metropolitan setting of Melbourne. The effectiveness of road lighting in reducing night accidents is widely recognised, with reported reductions ranging from 13% to 75% according to the CIE 008–1960 Street Lighting and Accidents report (2022). The study conducted by Yang et al. (2019) found a correlation between an increase in mean illuminance and a decrease in the incidence of nocturnal accident. It is worth mentioning that the presence of road lighting decreases both the quantity and intensity of accidents. Conversely, the lack of street lighting during nighttime has the most significant effect on fatalities and severe injuries (Yannis et al., 2013).

The presence of a positive coefficient for the spatial lag variable in our modelling result, reveals the existence of a spatial component in the frequency of car accident. The accident frequency in the present suburb is influenced by the accident frequencies observed in neighboring neighbourhoods with higher rates (Matkan et al., 2013; Saha et al., 2020). This observation is consistent with previous research. The presence of spatial autocorrelation has been observed in accidents occurring at adjacent locations (Liu, 2008). This highlights the interdependence of road safety in Melbourne’s urban environment, emphasizing the need to incorporate wider spatial patterns into traffic control systems. Integrating spatio-temporal dependence has been discovered to improve the predicted precision of accident models. Studies including spatial correlation have demonstrated a higher level of accuracy in their predictions (Aguero-Valverde and Jovanis, 2008). There is a notable presence of

spatial autocorrelation and spillover effects in the data related to the frequency of accident on motorways (Wen et al., 2019). According to Aguero-Valverde and Jovanis (2008), the typical effective distance across which spatial correlation occurs is approximately 168 m. The inclusion of a spatial element in models has shown to enhance the accuracy of accident count estimations (Liu, 2008).

The proximity to central business districts (CBDs) was a significant factor in the frequency of injurious accidents (Avrahamov-Kraft et al., 2022). It is worth mentioning that in Melbourne, a considerable number of accidents take place between 7:00 p.m. and 6:00 a.m. in the vicinity of hotels, clubs, and nightclubs (Toran Pour et al., 2017). According to Katrakazas et al. (2021), congested traffic is associated with a greater number of confrontations, and disputes are further intensified by larger variances in speed. The study conducted in Texas (Xu et al., 2014) found that speed restrictions in Central Business District (CBD) locations have a negative correlation with accident rates but a positive correlation with mortality rates. Work zones are also considered important indicators of accident (Eustace et al., 2015).

Our study showed a higher likelihood of accidents in proximity of urban parks and greenspaces. In similar, the study conducted by Luo et al. (2022) found that walkers near green spaces have higher road accidents in comparison to bicycles. Trees, street lighting, traffic lights, main avenues, and curves have all been linked to accidents in the environment (Chaparro et al., 2018).

Our model confirmed the critical situation of Central Melbourne in terms of accident frequency over a 20-year period. The commercial areas of Melbourne, characterized by a lively atmosphere and a high concentration of companies and workplaces, frequently encounter

increased traffic congestion during peak periods, hence amplifying the potential for accident (Oxley et al., 2013). There is a direct relationship between the number of accident and the presence of business and commercial areas, as observed in Honolulu, Hawaii (Kim et al., 2010).

5. Conclusions

Investigating road accident temporal and spatial patterns are essential for several reasons. It may help us build prevention strategies that target high-accident times and areas, distribute resources more efficiently, improve emergency response times, and assess road safety initiatives. The State and Local governments may promote safer roads and minimise road injuries and deaths by analysing GMA road accident patterns. These data may improve Victorian policy and support evidence-based actions for a safer, more resilient transportation infrastructure. The intricate interaction between built environment, social, and transportation variables affects accident risk in different places. Special land use activities such as commercial and parkland in addition to network provision, lightning employment density may affect road safety, and knowing how built environment variables contribute to accidents can help building safer roads and minimise injuries and deaths. The findings of this research may be used by urban planners to enhance land-use allocation and transportation planning, particularly in determining suitable locations for forthcoming developments, particularly those of significant size that may have substantial traffic risk implications. These insights, like symphony harmonics, increase our knowledge of road traffic accident and contribute to evidence-based policies that make roads safer. A comprehensive strategy that adeptly navigates the multiple cadences of a symphony devoted to creating a transport landscape differentiated by safety and sustainability is required to address the built environment's effect on road traffic accidents.

To mitigate road accident frequencies in Melbourne, Victoria, various measures can be enacted. One critical measure involves implementing lower speed limits in local streets with a priority of traffic calming in Central Melbourne and inner urban suburbs, resulting in modest reductions in mean speed and significant benefits for vehicles traveling at higher speeds (Corben, 2022). Moreover, designing the road system to manage accident and prevent fatalities and serious injuries is imperative, often accomplished through a blend of vehicle, infrastructure, and speed requirements as seen in the case study of Victoria, Australia (Truong et al., 2022).

While this study contributed to a growing body of research on the road safety, in Australia, it has some limitations. Due to computationally expensive datasets and intricate three-dimensional visualisations, space-time cube analysis may be challenging. Spatial and temporal resolutions may affect the detection of events across scales, and sparse data distribution may result in analytical gaps (Enguehard et al., 2013). Time intervals, geographical units, and context may limit findings (ESRI, 2021). Another approach includes conducting trend analyses for vehicle collisions at intersections, aiding in identifying accident types and conditions contributing to the highest accident rates (Khalifa et al., 2022). Moreover, this study used police-collected road accident reports as its source of data. Nonetheless, police accident data tend to underestimate accident, particularly for vulnerable road users (Dhondt et al., 2013; Soltani et al., 2024). A more valid source of information may be obtained by the linking of police-reported accident data with hospital admissions data.

Conducting a spatio-temporal analysis after segmenting traffic accidents based on various parameters, such as accident type, road user type, accident cause, and injury type, is a potential avenue for future research. This strategy could yield valuable insights into the fundamental factors contributing to various categories of accidents and their corresponding outcomes. Using panel data analysis to investigate the dynamic interaction of factors influencing the severity of accidents across distinct time intervals and geographic locations is an additional direction. Incorporating weather-related parameters, such as wind speed, humidity, precipitation, solar radiation, and temperature as

influential factors that are highly dependent on space and time, into the analysis is an enticing avenue for future research.

CRedit authorship contribution statement

Ali Soltani: Conceptualization, Data curation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Mohsen Roohani Qadikolaei:** Data curation, Investigation, Software, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare no conflict of interest.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.tranpol.2024.02.006>.

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